

Lab 5 - Bivariate Normal Distribution

Robert Thomas Koh

12/3/25

Part 1: Independent Standard Normals

```
set.seed(67)
```

```
# Question 1
```

```
dmnorm(c(0,0), mean=c(0,0), varcov=diag(2))
```

```
## [1] 0.1591549
```

```
# Question 2
```

```
dnorm(0)*dnorm(0)
```

```
## [1] 0.1591549
```

```
# Question 3
```

```
pmnorm(c(2,2), mean=c(0,0), varcov=diag(2)) - pmnorm(c(-2,2), mean=c(0,0), varcov=diag(2)) - pmnorm(c(2
```

```
## [1] 0.9110697
```

```
# Question 4
```

```
(pnorm(2)-pnorm(-2))*(pnorm(2)-pnorm(-2))
```

```
## [1] 0.9110697
```

```
# Question 5
```

```
sim=rnorm(1000, mean=c(0,0), varcov=diag(2))
```

```
x1_sim=sim[,1]
```

```
x2_sim=sim[,2]
```

```
sim_data=data.frame(x1=x1_sim, x2=x2_sim)
```

```
print(head(sim_data, 10))
```

```
##           x1           x2
```

```
## 1  1.2194019 -0.12440050
```

```
## 2 -0.9574989 -0.07916951
```

```
## 3 -1.1434113 -1.05718789
```

```
## 4 -0.3845343  0.60834117
```

```
## 5  1.6724665 -1.53729435
```

```
## 6 -1.3782919  0.49543114
```

```
## 7  0.9381498 -0.37573528
```

```
## 8  0.5064244 -0.70952189
```

```
## 9 -0.7139704  0.13159310
```

```
## 10 -0.5688982 -0.09853550
```

```
# Question 6
```

```
cor(x1_sim, x2_sim)
```

```
## [1] -0.03743436
```

```
# Question 7
```

```
cor1=cor.test(x1_sim, x2_sim)
```

```
print(cor1$p.value)
```

```
## [1] 0.2369231
```

```
#Fail to reject bc p>.05
```

```
# Question 8
```

```
# When Alpha is 5% you can expect 5% false positive, same logic applies.
```

Part 2: Correlated Bivariate Normal

```
rho_xy=-0.4
```

```
sigma_x=5
```

```
sigma_y=3
```

```
mu_x=2
```

```
mu_y=4
```

```
MeanVec=c(mu_x, mu_y)
```

```
VarCovMatrix=matrix(c(sigma_x^2, rho_xy*sigma_x*sigma_y,  
                      rho_xy*sigma_x*sigma_y, sigma_y^2),  
                    nrow=2, byrow=TRUE)
```

```
print(MeanVec)
```

```
## [1] 2 4
```

```
print(VarCovMatrix)
```

```
##      [,1] [,2]
```

```
## [1,] 25 -6
```

```
## [2,] -6 9
```

```
# Question 1
```

```
dmnorm(c(2,4), mean=MeanVec, varcov=VarCovMatrix)
```

```
## [1] 0.01157682
```

```
# Question 2
```

```
dmnorm(MeanVec, mean=MeanVec, varcov=VarCovMatrix)
```

```
## [1] 0.01157682
```

```
dmnorm(c(1,3), mean=MeanVec, varcov=VarCovMatrix)
```

```
## [1] 0.01025035
```

```
dmnorm(c(3,5), mean=MeanVec, varcov=VarCovMatrix)
```

```
## [1] 0.01025035
```

```
# The value is expected to be greatest at mean
```

```
# Question 3
```

```
prob_less=pmnorm(c(1,3), mean=MeanVec, varcov=VarCovMatrix)
```

```
print(prob_less)
```

```
## [1] 0.09586766
```

```
# Question 4
prob_x_less=pmnorm(c(1,Inf), mean=MeanVec, varcov=VarCovMatrix)
prob_y_less=pmnorm(c(Inf,3), mean=MeanVec, varcov=VarCovMatrix)
prob_or=prob_x_less + prob_y_less - prob_less
print(prob_or)
```

```
## [1] 0.694314
```

```
# Question 5
prob_y_less - prob_less
```

```
## [1] 0.2735737
```

```
# Question 6
sim2=rnorm(1000, mean=MeanVec, varcov=VarCovMatrix)
x_sim=sim2[,1]
y_sim=sim2[,2]
sim_data2=data.frame(x=x_sim, y=y_sim)
print(head(sim_data2, 10))
```

```
##           x           y
## 1  6.45875549 -1.5014986
## 2 -0.03150956  4.4881369
## 3  6.56486075  5.1682972
## 4 16.37980744  0.2055685
## 5 -4.13191409  4.5289888
## 6  6.36014342  4.8446698
## 7  1.55371294  0.2210044
## 8  2.10001530 -0.8698211
## 9  1.76726556  9.0530538
## 10 2.94957314  5.4687127
```

```
# Question 7
cor(x_sim, y_sim)
```

```
## [1] -0.3757214
```

```
# Question 8
cor2=cor.test(x_sim, y_sim)
print(cor2$p.value)
```

```
## [1] 7.004124e-35
```

```
# There is no coorelation as p is significantly less than .05.
```

```
# Part (a)
1 - 0.5 * (3/2) * (3 - 2)
```

```
## [1] 0.25
```

```
# Part (b)
3^2 * (1 - -0.5^2)
```

```
## [1] 11.25
```

```
# Part (c)
1 / sqrt(27/7)
```

```
## [1] 0.5091751
```